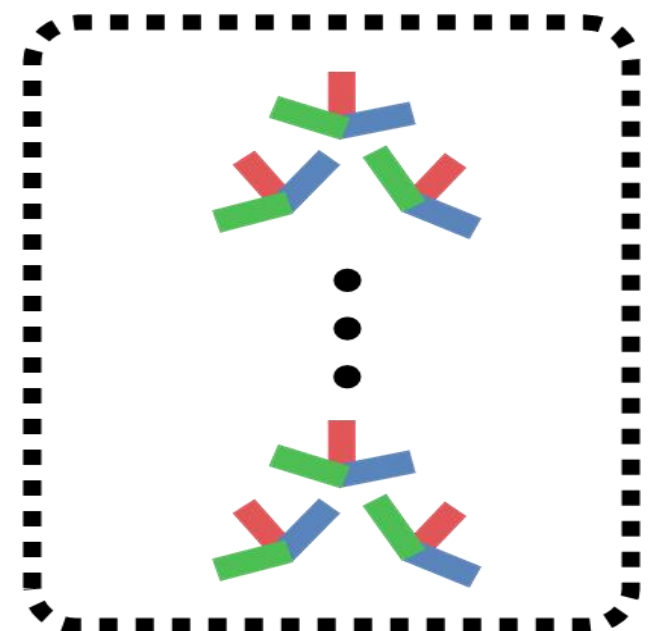
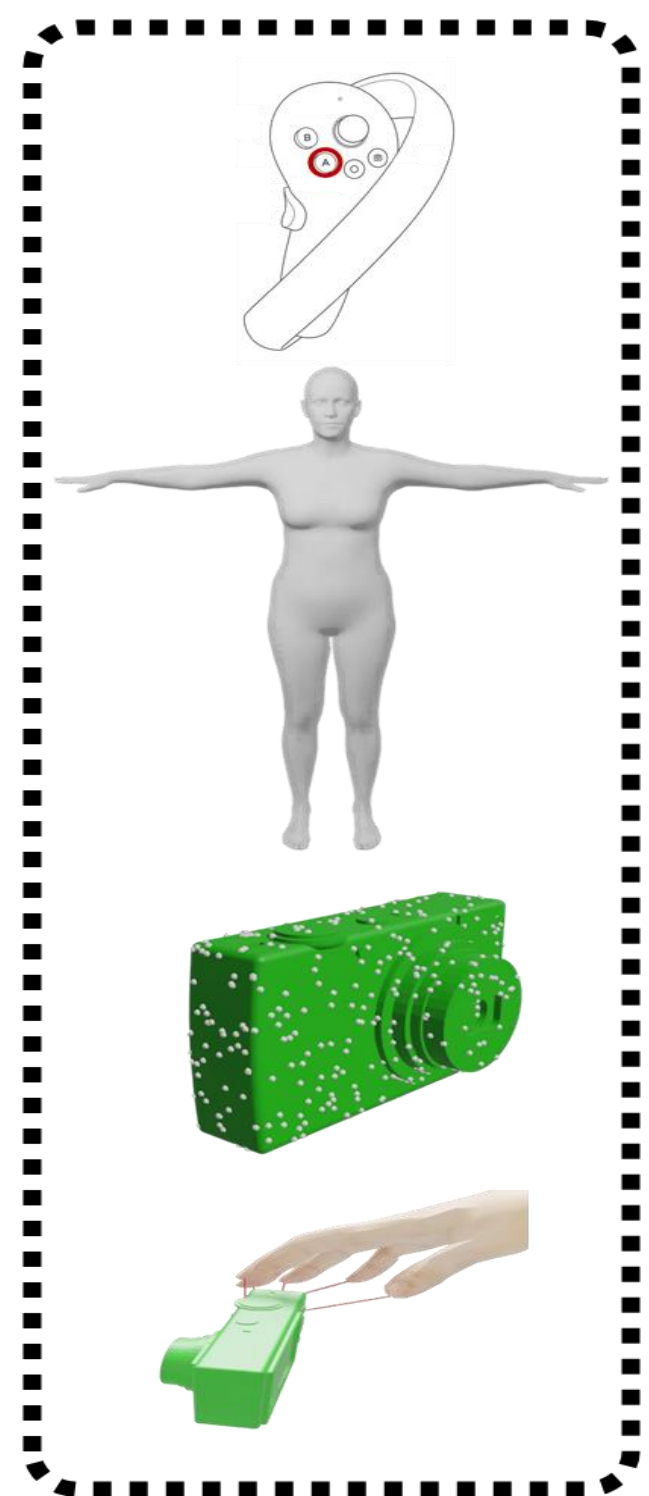


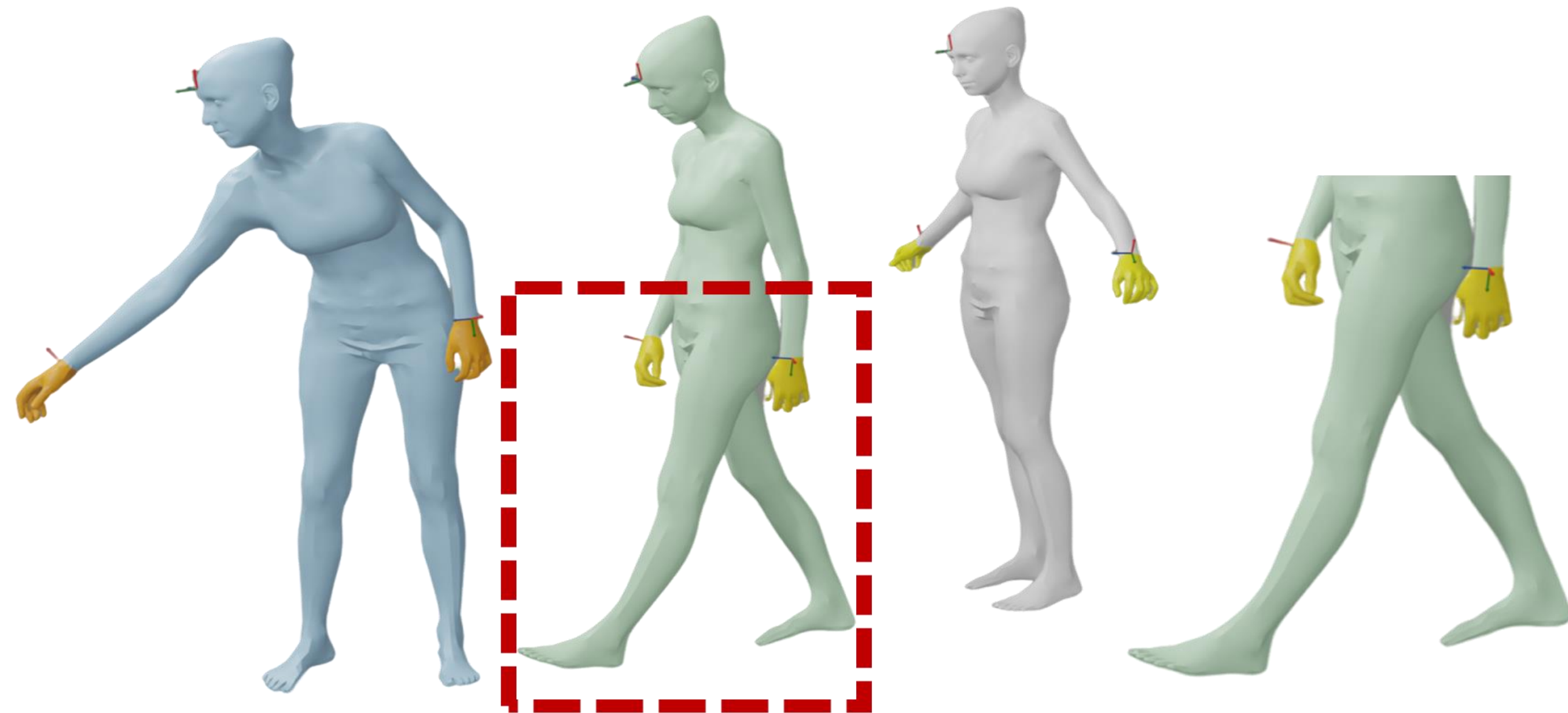
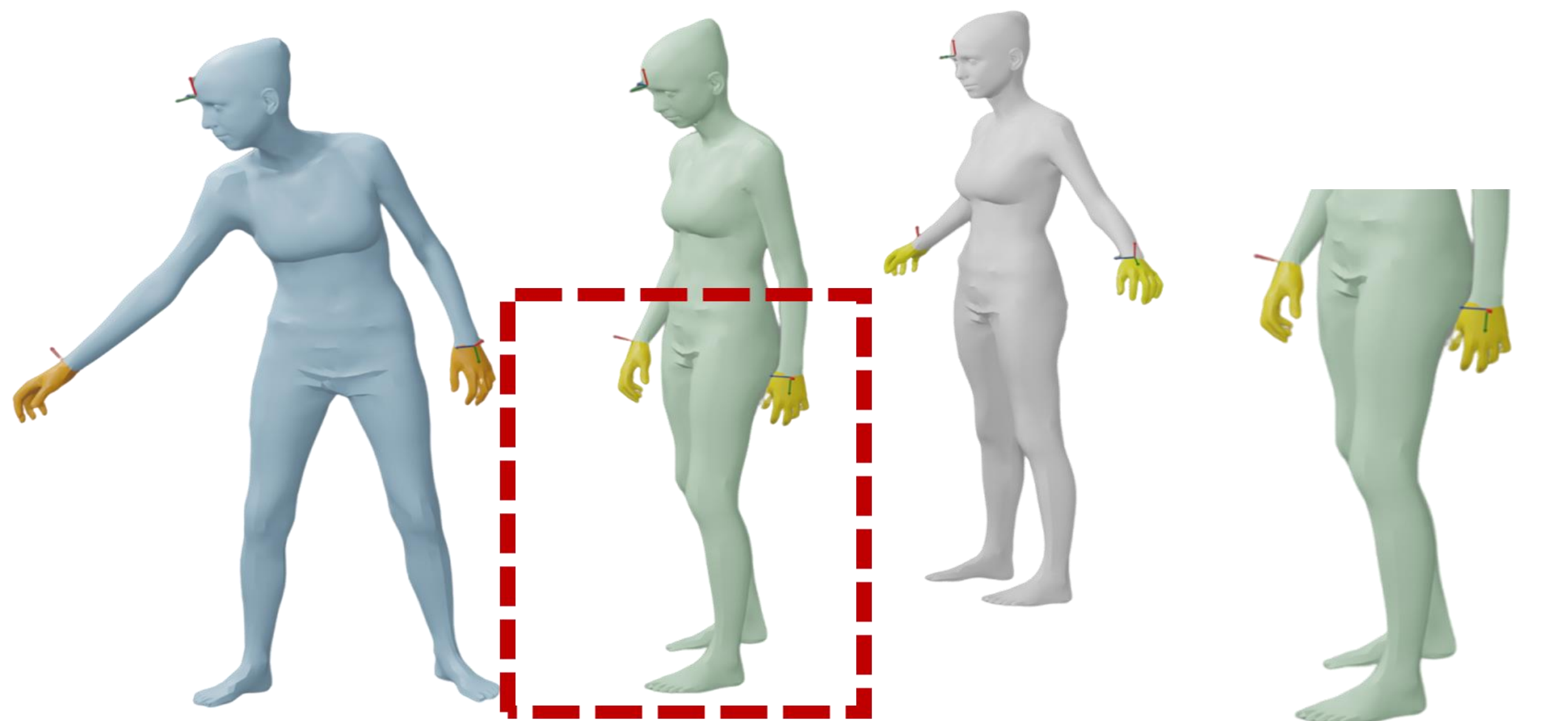


Sparse Tracking
Signals

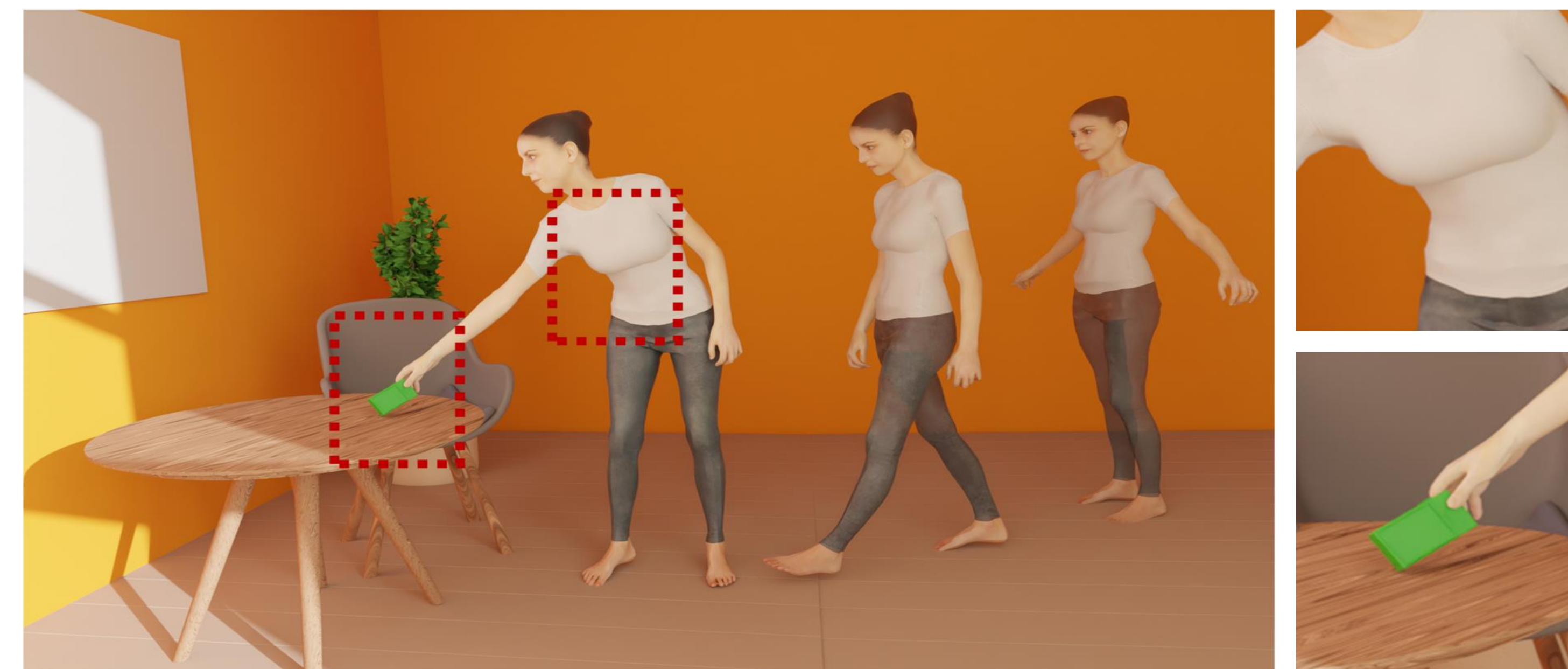


Object Manipulation
Representation

(a) Inputs



(b) Holistic Motion Reconstruction



(c) Generated Manipulation-aware Holistic Motion